

Free FRM Part I Formula Sheet 2026

Key Formulas for the Final Review

How to use this formula sheet

Use this as a final-review tool: recognize the formula, rewrite it from memory, then apply it immediately in practice questions. Do not use it as a replacement for the official curriculum or full practice exams.

1. Quantitative Analysis

2. Financial Markets & Products

3. Valuation & Risk Models

4. Do Not Confuse

1. Quantitative Analysis

Probability and conditional probability

Concept	Formula	How to use it / exam trap
Conditional probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$	Use when event B has already occurred. Trap: Do not use the unconditional probability of A.
Independent events	$P(A \cap B) = P(A)P(B)$	Use only when independence is given or justified. Trap: Mutually exclusive is not the same as independent.
Total probability	$P(A) = \sum_i P(A B_i)P(B_i)$	Breaks an event into mutually exclusive scenarios. Trap: The scenarios must cover the full sample space.
Bayes' theorem	$P(A B) = \frac{P(B A)P(A)}{P(B)}$	Use to reverse a conditional probability. Trap: Always compute the denominator consistently.
Union rule	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$	Use for probability of A or B. Trap: Subtract the overlap once.

Expected value, variance, and standardization

Concept	Formula	How to use it / exam trap
Expected value	$E[X] = \sum_i x_i p_i$	Weighted average of possible outcomes.
Variance	$\text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2$	Measures dispersion around the mean. Trap: Variance is in squared units.
Sample variance	$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$	Use n minus 1 for sample variance. Trap: Do not divide by n for sample variance.
Standard deviation	$\sigma = \sqrt{\text{Var}(X)}$	Same unit as the original variable.
Standard score	$z = \frac{x - \mu}{\sigma}$	Converts an observation into standard normal units.

Covariance, correlation, and portfolio risk

Concept	Formula	How to use it / exam trap
Covariance	$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$	Shows directional co-movement. Trap: Scale-dependent; hard to compare across pairs.
Correlation	$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$	Standardized co-movement between -1 and +1. Trap: Correlation is not causation.
Portfolio return	$E[R_p] = \sum_i w_i E[R_i]$	Weighted average expected return.
Two-asset variance	$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$	Correlation drives diversification benefit. Trap: Do not forget the cross-product term.
Matrix variance	$\sigma_p^2 = \sum_i \sum_j w_i w_j \text{Cov}(R_i, R_j) = w^T \Sigma w$	Compact multi-asset portfolio formula. Trap: Weights must match the covariance matrix order.

Regression and hypothesis testing

Concept	Formula	How to use it / exam trap
Simple regression	$Y_i = \alpha + \beta X_i + \varepsilon_i$	Models Y as a linear function of X plus error.
Regression beta	$\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)}$	Measures slope or sensitivity.
Test statistic	$t = \frac{\hat{\theta} - \theta_0}{SE(\hat{\theta})}$	Compare to critical value or use p-value. Trap: Check one-tailed versus two-tailed wording.
Confidence interval	$\hat{\theta} \pm z_{\alpha/2} SE(\hat{\theta})$	Use for large-sample normal approximation. Trap: For small samples, t critical values may apply.
Sharpe ratio	$SR = \frac{R_p - R_f}{\sigma_p}$	Excess return per unit of total risk. Trap: Uses total risk, not beta risk.

Distributions and VaR basics

Concept	Formula	How to use it / exam trap
Binomial probability	$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$	Use for k successes in n independent trials.
Normal-loss VaR	$VaR_c = \mu_L + z_c \sigma_L$	Loss quantile at confidence level c. Trap: Use loss distribution sign convention.
Return-based VaR	$VaR_c \approx W(z_c \sigma_R - \mu_R)$	Converts return volatility into currency loss. Trap: Mean is often ignored in short horizons.
Expected shortfall	$ES_c = E[L L > VaR_c]$	Average loss once VaR threshold is breached. Trap: ES is not the same as VaR.

2. Financial Markets and Products

Time value of money and interest rates

Concept	Formula	How to use it / exam trap
Present value	$PV = \frac{FV}{(1+r)^T}$	Discount future cash flows to today.
Future value	$FV = PV(1+r)^T$	Compound present value forward.
Effective annual rate	$EAR = \left(1 + \frac{r_m}{m}\right)^m - 1$	Converts m-times compounding to annual effective rate.
Continuous compounding	$FV = PVe^{rT}$	Use when rates are continuously compounded.
Discount factor	$DF(0, T) = e^{-rT}$	Used in bonds, forwards, and swaps.

Bonds, duration, and convexity

Concept	Formula	How to use it / exam trap
Bond price	$P = \sum_{t=1}^T \frac{CF_t}{(1+y)^t}$	Present value of coupons and principal.
Macaulay duration	$D_M = \frac{\sum_{t=1}^T tPV(CF_t)}{P}$	Weighted average timing of cash flows.
Modified duration	$D_{mod} = \frac{D_M}{1+y}$	Price sensitivity to yield changes. Trap: Adjust formula if compounding frequency differs.
Duration price effect	$\frac{\Delta P}{P} \approx -D_{mod}\Delta y$	First-order bond price approximation. Trap: Price moves opposite to yield.
Convexity adjustment	$\frac{\Delta P}{P} \approx -D_{mod}\Delta y + \frac{1}{2}C(\Delta y)^2$	Adds curvature for larger yield moves.

Forwards, futures, and swaps

Concept	Formula	How to use it / exam trap
Forward price, no income	$F_0 = S_0 e^{rT}$	Use when the underlying has no income.
Forward price, yield	$F_0 = S_0 e^{(r-q)T}$	Use when asset pays continuous yield q.
Known income	$F_0 = (S_0 - PV(I))e^{rT}$	Subtract PV of known income first.
Long forward value	$V_t = S_t - Ke^{-r(T-t)}$	Value after initiation. Trap: Forward price and forward value are different.
Swap fixed rate	$R_{swap} = \frac{1 - DF(0, T_n)}{\sum_{i=1}^n \Delta_i DF(0, T_i)}$	Fixed rate that sets initial swap value near zero.

Options and put-call parity

Concept	Formula	How to use it / exam trap
Call payoff	$\max(S_T - K, 0)$	Right to buy at strike K.
Put payoff	$\max(K - S_T, 0)$	Right to sell at strike K.
Put-call parity	$C + Ke^{-rT} = P + S_0$	Core no-arbitrage relation for European options. Trap: Use the correct discounting of K.
Black-Scholes call	$C = S_0 N(d_1) - Ke^{-rT} N(d_2)$	European call under BSM assumptions.
Black-Scholes put	$P = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$	European put under BSM assumptions.
BSM d terms	$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$	Know how each input affects option value.

3. Valuation and Risk Models

Value-at-Risk and expected shortfall

Concept	Formula	How to use it / exam trap
VaR loss quantile	$P(L > VaR_c) = 1 - c$	VaR is a threshold loss at confidence level c .
Normal-loss VaR	$VaR_c = \mu_L + z_c \sigma_L$	Use for normally distributed losses.
Return-based VaR	$VaR_c \approx W(z_c \sigma_R - \mu_R)$	Converts portfolio return risk into currency loss.
Expected shortfall	$ES_c = E[L L > VaR_c]$	Average tail loss beyond VaR.
Normal ES	$ES_c = \mu_L + \sigma_L \frac{\phi(z_c)}{1 - c}$	Tail mean under normal-loss assumption. Trap: Check whether c is 95%, 99%, etc.

Delta-normal and portfolio VaR

Concept	Formula	How to use it / exam trap
Portfolio sigma	$\sigma_p = \sqrt{w^T \Sigma w}$	Required before parametric portfolio VaR.
Parametric VaR	$VaR_c \approx W z_c \sigma_p$	Common approximation when mean is ignored.
Delta approximation	$\Delta V \approx \delta \Delta S$	First-order approximation for options/positions.
Delta-gamma	$\Delta V \approx \delta \Delta S + \frac{1}{2} \Gamma (\Delta S)^2$	Adds curvature for nonlinear instruments.
Component VaR	$VaR_p \approx \sum_i C VaR_i$	Allocates portfolio VaR across positions.

Volatility and correlation models

Concept	Formula	How to use it / exam trap
EWMA variance	$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$	Recent returns get more weight when lambda is lower.
GARCH(1,1)	$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$	Volatility depends on long-run term, shock, and persistence.
Volatility scaling	$\sigma_T = \sigma_1 \sqrt{T}$	Square-root-of-time approximation. Trap: Assumes independent returns and stable volatility.
Covariance from rho	$\text{Cov}(R_i, R_j) = \rho_{ij} \sigma_i \sigma_j$	Converts correlation into covariance.

Credit risk and performance measures

Concept	Formula	How to use it / exam trap
Expected loss	$EL = PD \times LGD \times EAD$	Core expected credit loss formula.
Loss given default	$LGD = 1 - RR$	Recovery reduces loss severity.
Unexpected loss	$UL \approx \sqrt{\text{Var}(L)}$	Represents uncertainty around expected loss.
CAPM return	$E[R_i] = R_f + \beta_i (E[R_m] - R_f)$	Required return for systematic risk.
Beta	$\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$	Sensitivity to market movements.
Jensen's alpha	$\alpha_i = R_i - [R_f + \beta_i (R_m - R_f)]$	Risk-adjusted excess return.

Option Greeks quick reference

Concept	Formula	How to use it / exam trap
Delta	$\Delta = \frac{\partial V}{\partial S}$	Sensitivity to underlying price.
Gamma	$\Gamma = \frac{\partial^2 V}{\partial S^2}$	Curvature with respect to underlying price.
Vega	$\nu = \frac{\partial V}{\partial \sigma}$	Sensitivity to volatility.
Theta	$\Theta = \frac{\partial V}{\partial t}$	Sensitivity to time decay.
Rho	$\rho = \frac{\partial V}{\partial r}$	Sensitivity to interest rates.

4. Do Not Confuse

Duration vs. Convexity

Duration	Convexity
First-order approximation of bond price sensitivity to yield changes. Best for small yield moves. $\frac{\Delta P}{P} \approx -D_{mod} \Delta y$	Second-order correction that captures curvature in the price-yield relationship. $\frac{\Delta P}{P} \approx -D_{mod} \Delta y + \frac{1}{2} C(\Delta y)^2$

VaR vs. Expected Shortfall

VaR	Expected Shortfall
Threshold loss at a chosen confidence level. It does not tell you the average size of losses beyond the threshold. $P(L > VaR_c) = 1 - c$	Average loss conditional on being beyond the VaR threshold. More informative about tail severity. $ES_c = E[L L > VaR_c]$

Correlation vs. Covariance

Correlation	Covariance
Standardized measure between -1 and +1. Easier to compare across asset pairs. $\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$	Scale-dependent co-movement measure. Useful inside covariance matrices. $\text{Cov}(X, Y) = \rho_{XY} \sigma_X \sigma_Y$

Forward Price vs. Forward Value

Forward price	Forward value
Delivery price that makes the contract value zero at inception. $F_0 = S_0 e^{rT}$	Current mark-to-market value of an existing forward contract. $V_t = S_t - K e^{-r(T-t)}$

Expected Loss vs. Unexpected Loss

Expected loss	Unexpected loss
Average credit loss expected over the horizon. Often priced into lending decisions. $EL = PD \times LGD \times EAD$	Uncertainty or volatility around expected loss. More relevant for capital and stress concerns. $UL \approx \sqrt{\text{Var}(L)}$

Alpha vs. Beta

Alpha	Beta
Risk-adjusted excess performance after accounting for market exposure. $\alpha_i = R_i - [R_f + \beta_i(R_m - R_f)]$	Sensitivity of an asset or portfolio to market movements. $\beta_i = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$

Arithmetic Return vs. Geometric Return

Arithmetic return	Geometric return
Simple average of periodic returns. Often higher when returns are volatile. $\bar{R} = \frac{1}{n} \sum_i R_i$	Compound average growth rate over time. Better reflects realized multi-period growth. $R_G = \left(\prod_{i=1}^n (1 + R_i) \right)^{1/n} - 1$

Disclaimer

This resource is intended as a focused revision aid for FRM Part I candidates. For complete coverage, candidates should refer to the official GARP curriculum and Learning Objectives and practice with realistic exam-style questions.